

Common Ownership Does Not Have Anti-competitive Effects in the Airline Industry

Dennis, Gerardi, and Schenone (2022)

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Background

- In 1980, 33% of publicly traded equity was traded by institutions. In 2018, that number had risen to 61%.
- Does this increase in *Common Institutional Ownership* suggest, or even necessitate, that these institutional owners will force these firms to behave in a more anti-competitive manor?

Research Question

Does *Common Ownership* cause anti-competitive behavior between firms, as measured by average fare price?

Previous Works

What relevant work was leveraged to break down this question:

- ① Azar, Schmalz, & Tecu (AST, 2018) - Found evidence for a casual relationship between *concentration of common ownership* and average ticket prices
- ② Hirschman (1970) - Identified two methods in which equity holder exercise control over firms; Voice and Exit¹
- ③ O'Brein, & Salop (2000) - Derived a theoretical measure of cross ownership, arising from manager profit maximization

¹Also considered more recently by Edmans (2009), Admati & Pfleiderer (2009), & Edmans & Manso (2011)

Importance of AST

- The AST results suggested a casual relationship between concentration of common ownership and average fare price
- If AST found these results, does this not answer our research question?
- First, the authors recreate the AST results, then they address where it falls short

Revisiting O'Brein & Salop

- O'Brein & Salop developed a measure of cross ownership that arises as a result of manager profit maximizing. Formally defined by:

$$HHI\Delta_{rt} = \sum_j \sum_{k \neq j} \left(\frac{\sum_i \gamma_{ijt} \beta_{ikt}}{\sum_i \gamma_{ijt} \beta_{ijt}} \right) s_{rjt} \cdot s_{rkt} \quad (1)$$

- Where γ_{ij} represents owner i 's control over carrier j in period t , and β_{ij} represents owner i 's cash-flow rights in carrier j . Both γ_{ij} and β_{ij} represent as fractions of total shares outstanding.

Construction of AST - Part 1

- The main econometric tool leveraged is a regression of the logarithm of average ticket prices, in market (route) r , during year-quarter t , on the measure of common ownership $HHI\Delta_{rt}$, with a set of control variables.

$$\log(p_{rjt}) = \alpha \cdot HHI\Delta_{rt} + \eta \cdot HHI_{rt} + \theta \cdot X_{rjt} + \alpha_t + v_{rj} + \varepsilon_{rjt}$$

- Standard-errors are double clustered by market-carrier and year-quarter.
- All regressions are weighted by the average number of passengers for the market and carrier over time.

Construction of AST - Part 2

- The second tool is a similar regression, rather done at the market level rather than the market-carrier level. Taking the form:

$$\log(p_{rt}) = \beta \cdot HH\Delta_{rt} + \gamma HHI_{rt} + \theta \cdot X_{rt} + \alpha_t + v_r + \varepsilon_{rt}$$

- Standard-errors are doubled clustered by market and year-quarter.
- Regression weights are given by average number of passengers in the market over time.

Construction of AST - Part 3

The control variables used in this regression are as follows:

- | | |
|-----------------------------------|---|
| ① Number of Nonstop Carriers | ⑦ Traveling Connect |
| ② Southwest Indicator | ⑧ Log(Population) |
| ③ Other LCC Indicator | ⑨ Log(Income per Capita) |
| ④ Share of Passengers | ⑩ Log(Distance) interacting Year-Quarter fixed effect |
| ⑤ Traveling Connect, Market-level | ⑪ Year-Quarter fixed effect |
| ⑥ Share of Passengers | ⑫ Market-Carrier fixed effect |

AST Results - Part 1

Panel A: Market-Carrier-Level Regressions						
	AST			DGS		
	(1)	(2)	(3)	(4)	(5)	(6)
$HHI\Delta$	0.194*** (0.046)	0.219*** (0.039)	0.149*** (0.038)	0.194*** (0.046)	0.218*** (0.039)	0.147*** (0.038)
HHI	0.221*** (0.025)	0.230*** (0.025)	0.165*** (0.021)	0.222*** (0.025)	0.230*** (0.025)	0.163*** (0.021)
Log(Distance) · Yr-Qtr FE		X	X		X	X
Yr-Qtr FE	X	X	X	X	X	X
Market-Carrier FE	X	X	X	X	X	X

Standard errors are in parentheses. ***, **, * indicate significance at the 1%, 5%, and 10% level, respectively.

AST Results - Part 2

Panel B: Market-Level Regressions						
	AST			DGS		
	(1)	(2)	(3)	(4)	(5)	(6)
$HHI\Delta$	0.325*** (0.045)	0.311*** (0.040)	0.202*** (0.036)	0.323*** (0.045)	0.344*** (0.041)	0.197*** (0.037)
HHI	0.365*** (0.032)	0.357*** (0.031)	0.256*** (0.024)	0.365*** (0.032)	0.365*** (0.033)	0.256*** (0.025)
Log(Distance) · Yr-Qtr FE		X	X		X	X
Yr-Qtr FE	X	X	X	X	X	X
Market FE	X	X	X	X	X	X

Standard errors are in parentheses. ***, **, * indicate significance at the 1%, 5%, and 10% level, respectively.

The Big Issue with AST

AST has one major issue we seek to address; AST does not take into consideration that the value of $HHI\Delta_{rt}$ is a composite of two values:

$$HHI\Delta_{rt} = \sum_j \sum_{k \neq j} \underbrace{\left(\frac{\sum_i \gamma_{ijt} \beta_{ikt}}{\sum_i \gamma_{ijt} \beta_{ijt}} \right)}_{\text{Ownership \& Control}} \underbrace{s_{rjt} \cdot s_{rkt}}_{\text{Market Share}} \quad (2)$$

- First, the effect of *common ownership & control*
- Second, the effects of *increased market share*

Improving the Measure

In this paper, the author makes the following postulation:

The correlation between concentration of common ownership and average ticket price is a result of larger market shares; Not a result of common ownership concentration.

In order to determine which factor is responsible for the positive and statistically significant coefficient estimate, they conduct a **Placebo Analysis**.

Constructing the Analysis - Part 1

- In order to run a placebo analysis, the author establishes two placebo variables.
- The first variable retains time-series variation in market-share but removes time-series variance in ownership & control.
- Thus, the authors fix γ_{ij} and β_{ij} (control and cash-flow rights respectively) to the first year-quarter of observation, for all year-quarters.
- This placebo variable is denoted,

$$HHI\Delta_{True\ MS}^{2001:Q1}, \dots, HHI\Delta_{True\ MS}^{2014:Q4}$$

Constructing the Analysis - Part 2

- Naturally, the second placebo variable needs to be the negation of the first variable.
- Thus, the second placebo retains any time-series variation in ownership & control while muting time-series variance in market-shares.
- Following a similar strategy, for all year-quarter samples they replace the value of airline j 's market share in route r with the value in the first sample period.
- The true ownership placebo variable will be denoted, $HHI\Delta_{True\ Own}^{2001:Q1}, \dots, HHI\Delta_{True\ Own}^{2014:Q4}$

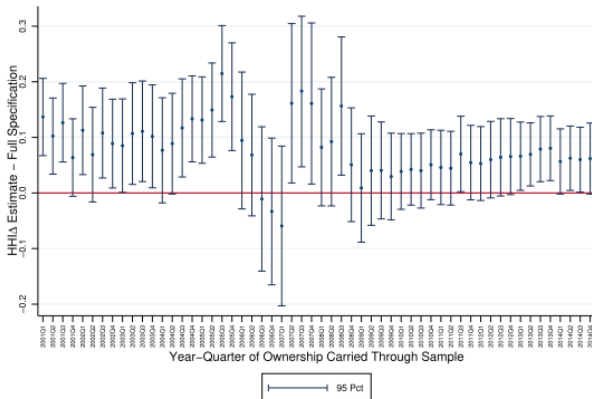
What Should We See

Before looking at the results, first we should know what to look for in the estimates.

- 1 If **concentration of common ownership** is responsible, we expect coefficient estimates for $HHI\Delta_{True\ MS}$ to be insignificant, and estimates for $HHI\Delta_{True\ Own}$ variable to be significant.
- 2 If **variations in market-share** are responsible, we expect the opposite, for coefficient estimates for $HHI\Delta_{True\ Own}$ to be insignificant, and estimates for $HHI\Delta_{True\ MS}$ to be significant.

Market-Carrier Results - Part 1

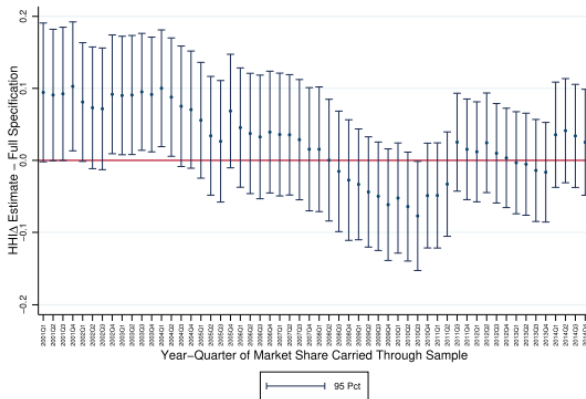
Panel (a) True Market Shares and Placebo Ownership



Market-Carrier Fixed Effects Regressions. Standard Errors Clustered at Market-Carrier and Date Level.

Market-Carrier Results - Part 2

Panel (b) True Ownership and Placebo Market Shares



Market-Carrier Fixed Effects Regressions. Standard Errors Clustered at Market-Carrier and Date Level.

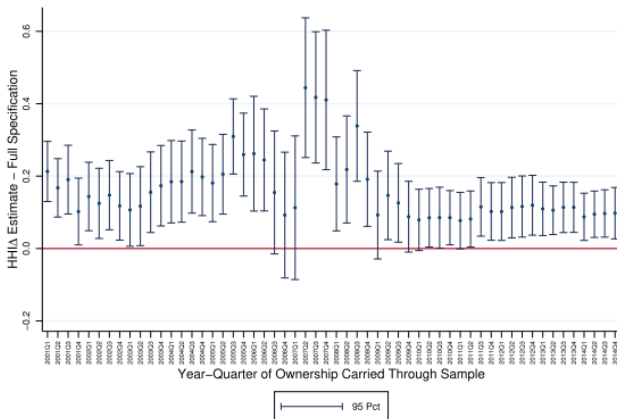
Market-Carrier Results - Part 3

Panel A: Market-Carrier-Level Regressions		
	True Market Shares Placebo Ownership/Control	Placebo Market Shares True Ownership/Control
	(1)	(2)
Mean of 56 $HHI\Delta$ estimates	0.081***	0.027
<i>Standard Error Computation:</i>		
Method 1	(0.014)	(0.032)
Method 2	(0.011)	(0.024)

Standard errors are in parentheses. ***, **, * indicate significance at the 1%, 5%, and 10% level, respectively.

Market Level Results - Part 1

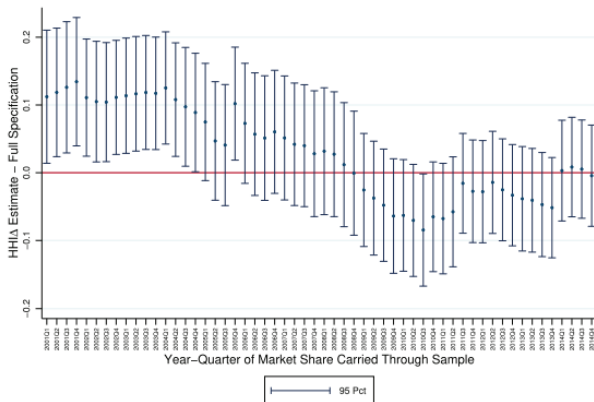
Panel (a) True Market Shares and Placebo Ownership



Market-Level Fixed Effects Regressions. Standard Errors Clustered at Market and Date Level.

Market Level Results - Part 2

Panel (b) True Ownership and Placebo Market Shares



Market Fixed Effects Regressions. Standard Errors Clustered at Market and Date Level.

Market Level Results - Part 3

Panel B: Market-Level Regressions		
	True Market Shares Placebo Ownership/Control	Placebo Market Shares True Ownership/Control
	(1)	(2)
Mean of 56 $HHI\Delta$ estimates	0.161***	0.029
<i>Standard Error Computation:</i>		
Method 1	(0.027)	(0.053)
Method 2	(0.016)	(0.030)

Standard errors are in parentheses. ***, **, * indicate significance at the 1%, 5%, and 10% level, respectively.

What Can We Conclude

- These results suggest a clear result, the relationship estimated in AST was a result of correlation rather than causation between market-share and ownership/control
- Are we sure about making such a clear delineation? Are there reasons to doubt these results?

Assumptions about Control

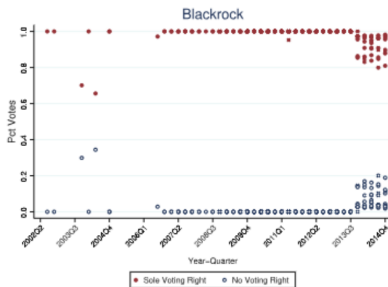
- A key element used in this paper is a measure of share control.
- Thus, a fundamental assumption is made when using this data. Specifically that this data is reported accurately and truly represents control and ownership over a firm.

How is Control Measured

- In this paper, control is measured using data from the SEC
- This data is self-reported and does not have rigid guidelines
- So, how do institutions report their voting designations

Voting Designation

- There are three designations:
 - 1 No Voting Rights
 - 2 Shared Voting Rights
 - 3 Sole Voting Rights
- We see examples of what we might declare inaccurate reporting, examples like Vanguard & StateStreet.



Analyzing Difference in Designation

- In order to further analyze the placebo results, the authors regress using different sections of voter designation
- More specifically, they run both market-carrier and market-level regressions again, instead with new specifications on control
- $HHI\Delta_{Sole + Shared}$ which is a combination of shares reported as sole voting rights, as well as share voting rights
- $HHI\Delta_{Exit} = HHI\Delta_{Sole + Shared} + No$ is constructed to capture the idea that anyone with ownership of equity can exercise the option to exit

Results of Designation - Part 1

Panel A: Market-Carrier Regressions

	$HHI\Delta_{Sole + Shared}$ (1)	$HHI\Delta_{Exit}$ (2)
$HHI\Delta$	0.147*** (0.038)	0.044 (0.031)
HHI	0.163*** (0.021)	0.138*** (0.021)

Results of Designation - Part 2

Panel B: Market-Level Regressions

	$HHI\Delta_{Sole + Shared}$ (1)	$HHI\Delta_{Exit}$ (2)
$HHI\Delta$	0.197*** (0.037)	0.106*** (0.034)
HHI	0.256*** (0.025)	0.236*** (0.025)

What can We Conclude

- These are interesting results since voice and exit are considered to be complimentary mechanisms in the literature, see Levit (2019)
- Dasgupta & Piacetino (2015) show managers only listen to shareholder with the threat of exit
- McCahery, Sautner, & Starks (2008) show that these theoretical findings are also consistent with survey findings
- This gives significant reason to doubt our data on voting right designations

Putting It All Together

- We've seen that, while AST's results are certainly not representative, however the results of this analysis have also been called into question
- Hence, to build on this research, the *most important question* to ask is how can institutional control be measured more accurately
- Furthermore, theoretical research hoping to build in this area may attempt to construct a new measure of institutional control, that can leverage different data

Concluding Thoughts

- This analysis has built on an exceptional model for empirically assessing the effects of common ownership, that is also backed-up by a fairly robust theoretical measure
- However, there is still work to be done in both areas to assemble an analysis that may inform policy decisions

Thank You!

Questions?